REVISED VERSION · 16.10.2026

Machine Learning in financial, stock-market and political predictions

Which type of learning and which model to choose — a revised version of your two texts: the split of the course topics into supervised and unsupervised learning, and the note “ML_Trading”.

PARTS 1–2
Concepts and course topics: what is supervised, what is unsupervised, where deep learning fits.

PARTS 3–7
The revised text on predictions, with two new chapters (5 and 6) that were missing in the original.

AT THE END
List of changes compared with the original and the sources the corrections are based on.

1. First the concepts: the type of learning is not the type of model

Supervised learning — the data have a label, i.e. a target variable, and the model learns to predict it. If the label is a number, it is regression (e.g. return, spending); if it is a class — classification (e.g. up/down, will click / won’t click).

Unsupervised learning — there is no label. The algorithm looks for structure in the data itself: groups (clustering), directions of greatest variance (PCA) or unusual observations (anomalies).

Self-supervised learning — the label is created from the data itself, e.g. the next word in a text. This is how Word2Vec and language models learn; they are usually counted among the methods without manual labels.

Deep learning is not another type of learning but a family of models — neural networks. A network can learn supervised (an LSTM predicting a return), unsupervised (an autoencoder) or self-supervised (a language model).

Model family × type of learning — examples

MODEL FAMILY	SUPERVISED LEARNING — THERE IS A LABEL	NO LABEL — UNSUPERVISED / SELF-SUPERVISED
Linear and statistical models	linear regression, logistic regression, Ridge, Lasso	PCA
Geometric methods (distance, margin)	KNN, SVM	K-Means, One-Class SVM
Trees and their ensembles	decision tree, Random Forest, XGBoost, LightGBM	Isolation Forest (anomalies)
Neural networks (deep learning)	MLP, LSTM and Transformer for forecasts, sentiment classification	autoencoders; Word2Vec and language models (self-supervised)

THE KEY CORRECTION

In the original text deep learning stood next to supervised and unsupervised as a third option. That is a conceptual error: an LSTM or Transformer that predicts a price learns supervised — just like XGBoost. The proper comparison is **simple models (linear, trees) versus neural networks**, all within supervised learning.

2. Course topics: supervised or unsupervised?

The corrected split of the topics from your list. Where I know your course notebook, I give the label it uses.

TOPIC	TYPE OF LEARNING	WHY / WHAT TO WATCH OUT FOR
1-Linear_Regression	**supervised** regression	The label is a number — in the project the customer's yearly spending (`Yearly Amount Spent`).
2-Logistic_Regression	**supervised** classification	The label is a class — a click on an ad (`Clicked on Ad`).
3-K_Nearest_Neighbors	**supervised** classification (usually)	Needs labels (`TARGET CLASS`). Not to be confused with K-Means. Without labels only the nearest-neighbour search itself works (`NearestNeighbors`).
4-Decision_Trees_und_ Random_Forests	**supervised**	Label: whether the loan was not fully repaid (`not.fully.paid`). Exception without labels: Isolation Forest for anomaly detection.
5-Support_Vector_Machines	**supervised** usually	Label: the iris species. Exception without labels: One-Class SVM for anomaly detection.
6-K_Means_Clustering	**unsupervised** clustering	Uses no labels; in the course the `Private` column is only used for evaluation. Cluster numbers are assigned randomly.
7-Principal_Component_ Analysis	**unsupervised** dimensionality reduction	Looks for the directions of greatest variance in the data, not for groups. A tool for data preparation and visualisation, not a predictive model.
8-Recommender_Systems	**separate category**	Similarity computed from rating correlations has no target variable (closer to unsupervised). Matrix factorisation (SVD) learns to predict known ratings and is evaluated with RMSE on held-out ratings — in practice that is supervised learning. Recommendations based on product features (content-based) are supervised too.
9-Natural_Language_ Processing	**in the course: supervised**	Yelp project: the label is the number of stars (1 vs 5), model `MultinomialNB` , evaluation with a confusion matrix. In general NLP covers all types: supervised (sentiment, NER), unsupervised (topic modelling, text clustering) and self-supervised (Word2Vec, GloVe, language models).

The Keras module (deep learning, LendingClub project) is supervised learning — label `loan_repaid` . The simplest proof that deep learning is a type of model, not a type of learning.

3. Financial, stock-market and political predictions: what to choose?

There is no single best approach — it depends on the data and the task. The order of decisions, however, is simple: (1) if you want to predict something and you have a label, it is a task for supervised learning; (2) only then do you choose the model, starting with a simple one. On financial data simple models often win, and neural networks are not automatically better.

3.1 Type of learning

APPROACH	WHAT IT IS SUITED FOR	ADVANTAGES	DISADVANTAGES	WHEN TO CHOOSE
Supervised	Predicting the direction of prices and returns, election results, economic indicators — wherever there is a target variable	Optimises exactly what you want to predict; the result can be measured on test data and compared with a baseline	Needs a well-defined label and good features; overfitting and data leakage come easily, especially on financial data	Always when the goal is prediction — this is the starting point
Unsupervised K-Means, PCA, anomaly detection, topic modelling	Detecting market regimes, grouping voters, markets and texts, reducing the number of features, anomaly detection	Needs no labels; discovers the structure of the data	Predicts nothing by itself; the result has to be interpreted (cluster numbers are random); quality is hard to assess	As a complement: feature preparation, regime filter, data checks

3.2 Type of model (in supervised tasks)

MODEL	ADVANTAGES	DISADVANTAGES	WHEN TO CHOOSE
Linear models linear and logistic regression, Ridge, Lasso	Few parameters, so the lowest risk of overfitting with strong noise; the easiest to interpret (coefficients)	Do not capture non-linearities and interactions unless you add them as features	First benchmark; little data, a lot of noise
Trees and boosting Random Forest, XGBoost, LightGBM	Very strong on tabular data — on medium-sized datasets (approx. 10k rows) usually better than neural networks; capture non-linearities and interactions	Overfit easily (deep trees); interpretation needs tools (SHAP, permutation importance)	Tabular data with many features; the step after the linear model
Neural networks MLP, LSTM, Transformer, NLP models	Can capture complex temporal and text patterns; the best choice for text analysis (news, speeches)	Need a lot of data, very prone to overfitting, a black box, expensive to train	A lot of data (e.g. thousands of instruments, a long history) or text; when simpler models have reached their limit

4. What works in practice?

Stock market and finance

- On tabular data (technical and fundamental features) tree-based models do well, especially gradient boosting. The first point of reference, however, should be a linear model and a simple baseline.
- Neural networks (LSTM, Transformer) on a single instrument with a limited history often lose to simpler models, because financial data are noisy and non-stationary. In time-series forecasting benchmarks a simple linear model beat complex Transformers on all nine datasets (Zeng et al., 2023).
- With large datasets the result can be different: in a study of US stocks (thousands of shares, several decades) trees and neural networks beat linear models. Shallow networks were enough, though — the best one had 3 hidden layers — and the monthly R^2 was only 0.33–0.40% (Gu, Kelly, Xiu, 2020).
- Unsupervised learning is useful for detecting regimes (e.g. trending vs sideways market) — provided that you fit the regime model on past data only.

- Realistic expectations: the predictability of markets is small, and the edge after transaction costs is even smaller.

Political predictions

- There are few elections, so there are few labels too. Professional forecasts rely mainly on statistically combining many polls and "fundamental" data (economy, previous results), not on large ML models.
- Supervised learning makes sense where there are many observations: at the level of districts or regions, from polling, demographic and historical results data.
- NLP (sentiment from social media, speeches, news) can be an additional feature, but as a stand-alone way of predicting elections it has a poor track record — a review of studies called Twitter's predictive power "greatly exaggerated" (Gayo-Avello, 2012).
- Unsupervised learning: grouping topics, segmenting voters.

General rules

- Do you have a label (election result, price direction, up/down)? → supervised learning. Start with a simple model and a comparison against a baseline.
- Do you want to understand the structure of data without labels? → unsupervised learning.
- Test neural networks only when you have a lot of data and simpler models have reached their limit.

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5. Before you trust a result

This chapter was missing in the original. In financial predictions the way the model is tested decides the result more often than the algorithm itself.

- **Validation without data leakage.** Split the data chronologically and test with walk-forward. For tuning parameters use cross-validation with purging (purged k-fold) and a gap (embargo) between the sets (López de Prado, 2018). Fit everything that adapts to the data — scaling, PCA, the regime model — on the past only.
- **Baseline.** Compare with the simplest answer: "always up" or "always the majority class". The S&P 500 rose on 53.7% of trading days in 1951–2025, so a 55% directional accuracy is not much more than this simple rule.
- **Label definition.** Fix the horizon, the move threshold and what counts as success — e.g. the triple-barrier method: take profit, stop loss and a time limit (López de Prado, 2018).
- **Costs.** The goal is the result after fees and slippage, not the hit rate.
- **Backtest overfitting.** The more variants you test, the easier it is to get an accidentally good result. The Deflated Sharpe Ratio (Bailey, López de Prado, 2014), permutation tests and a parameter sensitivity analysis help.
- **With imbalanced classes accuracy is not enough.** Look at the confusion matrix and at the recall and precision of the minority class. An example from the course: the random forest had 84.55% accuracy against a baseline of 84.59% and detected 2% of the unpaid loans.

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6. Use in a rule-based strategy (e.g. Donchian breakouts)

- **Meta-labeling.** The rule generates a signal, and a supervised model judges whether to take it and how large a position to open. Label: whether the trade was profitable. This is usually more realistic than predicting the price itself (López de Prado, 2018).
- **Regime filter.** An unsupervised model (HMM, Gaussian mixture, K-Means) can replace a rule-based sideways-market filter. Fit it on a rolling window of past data only, and interpret the states or clusters yourself — their numbers are random.

7. A realistic summary

TASK	TO START	WHAT TO WATCH OUT FOR
Stock prediction / trading	Supervised: a linear model as a benchmark, then boosting (XGBoost, LightGBM)	Walk-forward, baseline, costs; with a rule-based strategy — meta-labeling
Financial predictions (macro, indicators)	Supervised: statistical or linear models, then trees	Monthly or quarterly data means few observations — simple models
Political predictions	Statistical combination of polls + supervised models at regional level; NLP as an additional feature	Few elections = few labels
Regime / anomaly detection	Unsupervised: HMM, Gaussian mixtures, K-Means, Isolation Forest	Only on past data; the states need interpreting
Analysis of large amounts of text (news, social media)	Neural networks / NLP models (e.g. transformer models for sentiment)	The result (sentiment) is usually a feature for a further supervised model

CONCLUSION

For most financial and political predictions the starting point is supervised learning — beginning with a simple model and a comparison against a baseline. Neural networks are not "better" by definition: with a small amount of noisy data they often do worse or only slightly better, at a higher risk of overfitting and a higher cost; they have the edge with large datasets and with text. Treat unsupervised learning as a supporting tool (regimes, features, anomalies), not as the main prediction engine. Whatever the model, the result depends on sound validation: without data leakage, with a baseline and with costs.

A. What I corrected compared with the original

Split of the course topics (screenshot)

1. NLP: in the course it is supervised learning (Yelp project with the number of stars as the label), not "partly unsupervised".
2. Recommender systems: instead of "the basic approaches are unsupervised" — a separate category; matrix factorisation predicts known ratings, so in practice it is supervised.
3. PCA: looks for the directions of greatest variance, not for groups; it is a tool for data preparation.
4. Added: KNN is not K-Means; exceptions without labels (One-Class SVM, Isolation Forest); the Keras module as an example that deep learning can be supervised.

Text on predictions (ML_Trading.pdf)

5. Separated the type of learning from the type of model — deep learning is a family of models, not an alternative to supervised (chapter 1 and table 3.2).
6. Removed the claim that supervised has a "lower tendency to overfit". Overfitting depends on model complexity and the amount of data — in the course an unrestricted decision tree had 73% accuracy against a baseline of 84.6%.
7. Interpretation: easy for linear models, but not for random forests and boosting.

8. Clarified when neural networks lose (little data, a single instrument) and when they win (large datasets, text) — with research results.
9. Political forecasts: few labels, the role of combining polls, the weak predictive value of social-media sentiment.
10. Added chapters 5 and 6: validation without leakage, baseline, label definition, costs, backtest overfitting; meta-labeling and the regime filter.

B. Sources

- Gu, S., Kelly, B., Xiu, D. (2020). *Empirical Asset Pricing via Machine Learning.* Review of Financial Studies; NBER Working Paper w25398. nber.org/papers/w25398.pdf
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- Grinsztajn, L., Oyallon, E., Varoquaux, G. (2022). *Why do tree-based models still outperform deep learning on typical tabular data?* NeurIPS. arxiv.org/abs/2207.08815
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- López de Prado, M. (2018). *Advances in Financial Machine Learning.* Wiley — purged k-fold validation with embargo, the triple-barrier method, meta-labeling.
- Bailey, D. H., López de Prado, M. (2014). *The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality.* Journal of Portfolio Management.
- Course results (random forest, decision tree, Yelp and Keras projects) — from your notebooks, described in detail in the PDF "How to read ML model results".